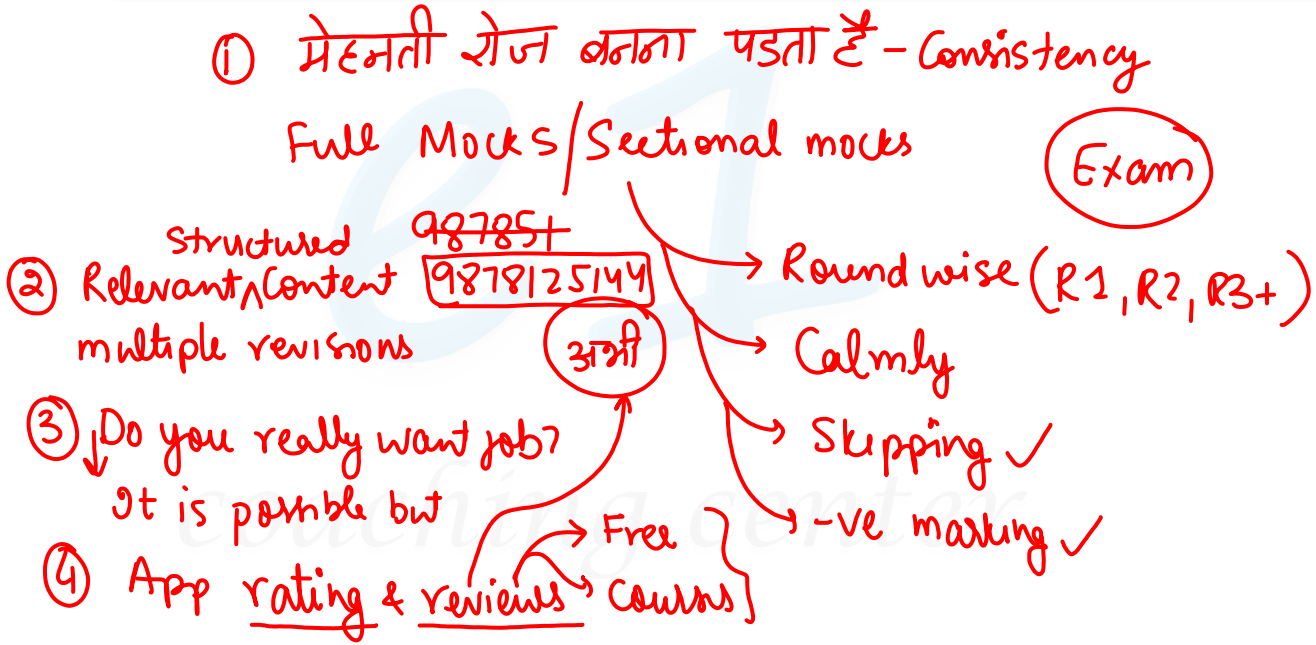
2 variables cube identities:



Pin Group

Web Version

$(a+b)^3$	$= a^3 + b^3 + 3ab(a+b)$
	$= a^3 + 3a^2b + 3ab^2 + b^3$
$(a-b)^3$	$= a^3 - b^3 - 3ab(a-b)$
	$= a^3 - 3a^2b + 3ab^2 - b^3$

$a^3 + b^3$	$= (a+b)^3 - 3ab(a+b)$
	$= (a+b)(a^2 - ab + b^2)$
$a^3 - b^3$	$= (a-b)^3 + 3ab(a-b)$
	$= (a-b)(a^2 + ab + b^2)$

HW

① If $p - 2q = 4$, then the value of $p^3 - 8q^3 - 24pq - 64$ is

अगर $p - 2q = 4$, तो $p^3 - 8q^3 - 24pq - 64$ का मान:

a) 2 b) 0 c) 3 d) -1

① put $q = 0$ $\Rightarrow p = 4$ put values here

---

② $(p-2q)^3 = 4^3$

$\Rightarrow p^3 - 8q^3 - 3p2q(p-2q) = 64$ (4)

$\Rightarrow p^3 - 8q^3 - 24pq - 64 = 0$

coaching center

② If $p = 999$, then the value of $\sqrt[3]{p(p^2 + 3p + 3) + 1}$ is

अगर $p = 999$, तो $\sqrt[3]{p(p^2 + 3p + 3) + 1}$ का मान:

a) 1000 b) 999 c) 998

d) 1002

$$\sqrt[3]{(p+1)^3}$$

$$= (p+1)$$

e1 coaching center

③ If $p = 99$, then value of $p(p^2 + 3p + 3)$ is

अगर $p = 99$, तो $p(p^2 + 3p + 3)$ का मान:

a) 999 b) 9999 c) 99999

d) 999999

$= (P+1)^3 - 1$

$= (100)^3 - 1$

$= 1000000 - 1$

$= 999999$

e1 coaching center

(4) If $p^3 + 3p^2 + 3p = 26$, then the value of $p^2 + 2p$ is:

यदि $p^3 + 3p^2 + 3p = 26$ है, तो $p^2 + 2p$ का मान ज्ञात कीजिए|

a) 8 b) 12 c) 10 d)15

$(p+1)^3 = 3^3$

$p+1 = 3$

$p = 2$

e1 coaching center

(5) If $m = -9, n = 8$, then the value of $m^3 - 3m^2 + 3m + 3n + 3n^2 + n^3$ is

अगर $m = -9, n = 8$, तो $\underbrace{m^3 - 3m^2 + 3m}_{-1} + \underbrace{3n + 3n^2 + n^3}_{+1}$ का मान:

a) 271 b) -271 c) -1 d) 0

$$m^3 - 3m^2(1) + 3m(-1)^2 - (1)^3$$

$$= (m-1)^3 + (n+1)^3$$

$$= -1000 + 729$$

$$= -271$$

coaching center

Hw

⑥ If $8x^3 - 27y^3 = (Ax + By)(Cx^2 - Dy^2 + 6xy)$, then $(A + B + C - D)$ is equal to: $(2x-3y)(4x^2+9y^2+6xy)$

यदि $8x^3 - 27y^3 = (Ax + By)(Cx^2 - Dy^2 + 6xy)$ है, तो $(A + B + C - D)$ निम्नलिखित में से किसके बराबर है?

a) -12 b) 12 c) 15 d) 9

$(2x)^3 - (3y)^3 =$

$a^3 - b^3 = (a-b)(a^2+b^2+ab)$

on comparing

A=2

B=-3

C=4

D=-9

$2 - 3 + 4 + 9 = 12$

coaching center

(7) If $27x^3 - 64y^3 = (Ax + By)(Cx^2 - Dy^2 + 12xy)$, then the value of $4A + B + 3C + 2D$ is:

HW

यदि $27x^3 - 64y^3 = (Ax + By)(Cx^2 - Dy^2 + 12xy)$ है, तो $4A + B + 3C + 2D$ का मान कितना होगा?

a) 5 b) 3 c) -3 d) -4

$(3x)^3 - (4y)^3 = (3x - 4y)(9x^2 + 16y^2 + 12xy)$

On comparing,

$A = 3,\ B = -4,\ C = 9,\ D = -16$

Ans $= 12 - 4 + 27 - 32 = 3$

$12 - 4 = 8$, $27 - 32 = -5$

coachingcenter

(8) HW

If $24\sqrt{3}x^3 + 5\sqrt{5}y^3 = (2\sqrt{3}x + \sqrt{5}y) \times (Ax^2 + Bxy + Cy^2)$, then what is the value of $(A^2 - B^2 + C^2)$?

यदि $24\sqrt{3}x^3 + 5\sqrt{5}y^3 = (2\sqrt{3}x + \sqrt{5}y) \times (Ax^2 + Bxy + Cy^2)$ है, तो $(A^2 - B^2 + C^2)$ का मान क्या होगा ?

a)108 b)128 c)109 d)139

$(12x^2 - 2\sqrt{15}xy + 5y^2)$

$A = 12,\ B = -2\sqrt{15},\ C = 5$

$8 \times 3\sqrt{3}$

$2^3 \quad (\sqrt{3})^3$

Ans = $144 - 60 + 25$

$= 84 + 25$

$= 109$

coaching center

⑨ If $250\sqrt{2}x^3 - 5\sqrt{5}y^3 = (5\sqrt{2}x - \sqrt{5}y)(Ax^2 + Bxy + Cy^2)$, then the value of $(A + C - \sqrt{10}B)$ is:

यदि $250\sqrt{2}x^3 - 5\sqrt{5}y^3 = (5\sqrt{2}x - \sqrt{5}y)(Ax^2 + Bxy + Cy^2)$ है, तो $(A + C - \sqrt{10}B)$ का मान है:

a) 10 b) 5 c) $5\sqrt{2}$ d) $2\sqrt{5}$

a b $50\ 5\sqrt{10}\ 5$

$50 + 5 - 50$

coaching center

(10) If $x^6 - 6\sqrt{6}y^6 = (x^2 + Ay^2)(x^4 + Bx^2y^2 + Cy^4)$, then what will be value of $(A^2 - B^2 + C^2)$?

HW

यदि $x^6 - 6\sqrt{6}y^6 = (x^2 + Ay^2)(x^4 + Bx^2y^2 + Cy^4)$ है, तो $(A^2 - B^2 + C^2)$ का मान ज्ञात करें|

a) 27 b) 42 ~~c) 36~~ d) 18

$(x^2)^3 - (\sqrt{6}y^2)^3 = (x^2 - \sqrt{6}y^2)(x^4 + \sqrt{6}x^2y^2 + 6y^4)$

On comparing, $A = -\sqrt{6}$, $B = \sqrt{6}$, $C = 6$

$\Rightarrow$ Ans $= 6 - 6 + 36 = 36$

coaching center

(11) If $8(x+y)^3-27(x-y)^3=(5y-x)(Ax^2+By^2+Cxy)$, then what is the value of $(A+B-C)$?

यदि $8(x+y)^3-27(x-y)^3=(5y-x)(Ax^2+By^2+Cxy)$ है, तो $(A+B-C)$ का मान ज्ञात करें|

a) 36 b) 16 c) -26 d) -16

$(A+B-C)$

$19x^2+7y^2-10xy$

$19+7+10=36$

(1) $[2(x+y)]^3-[3(x-y)]^3=$

a b

$(a-b)(a^2+b^2+ab)$

$4x^2+4y^2+8xy$

$9x^2+9y^2-18xy$

$6x^2-6y^2$

(2) $x=1, y=-1$

$+27\times 8=+6$ (□)

9 4

$64=4\times(26+C)$

16 -10

$8+27=-1\times A$

$19=A$

(3)

$x=1, y=0$,

$A=19$

$x=0, y=1$

$B=7$

$x=1, y=1$

$C=-10$

$A+B+C$

$8+27=$

$5\times B$

coaching center

(12) Given that $(2x+y)^3-(x+2y)^3=(x-y)\left[A\left(x^2+y^2\right)+Bxy\right]$, the value of $(2A-B)$ is:

दिया गया है कि $(2x+y)^3-(x+2y)^3=(x-y)\left[A\left(x^2+y^2\right)+Bxy\right]$ है, तो $(2A-B)$ का मान ज्ञात करें|

a) 7 b) 6 c) 0 d) 1

$x=1$

$y=-1$

$1-1+1=2(\square)$

coaching center

(13) If $(x+2y)^3+(2x-y)^3=(3x+y)\left[Ax^2+By^2+Cxy\right]$, then the value of $(3A-B-C):(A+B+C)$ is:

यदि $(x+2y)^3+(2x-y)^3=(3x+y)\left[Ax^2+By^2+Cxy\right]$ है, तो $(3A-B-C):(A+B+C)$ का मान क्या है?

a) 2 : 5 b) 3 : 4 c) 5 : 7 d) 1 : 7

$x=1, y=0$, $A=3$

$3 \cancel{1+8}= 3[A]$

$x=0, y=1$, $B=7$

$8 \cancel{+1} = B$

$x=1, y=1$

$C=-3$

$\cancel{27+1}= 4[10+C]$

7

$\frac{9-7+3}{3+7-3} = \frac{5}{7}$

coaching center

(14) $\frac{775\times775\times775+225\times225\times225}{77.5\times77.5+22.5\times22.5-77.5\times22.5}$ is equal to:

$\frac{775\times775\times775+225\times225\times225}{77.5\times77.5+22.5\times22.5-77.5\times22.5}$ का मान ज्ञात करें|

a) 100 b) 10000 c) 100000 d) 10

$100\times\frac{(a+b)(a^2+b^2-ab)}{a^2+b^2-ab}$

$=100\times1000=$

$\frac{5\times100}{3.53}$

coaching center

15 $\dfrac{(253)^3+(247)^3}{25.3\times2.53-62.491+2.47\times24.7} = 50\times10^k,\ k=?$

a) 3 b) 4 c) 2 d) -3

$1000\times500 = 50\times10^k$

$10^4 = 10^k$

coaching center

16 Simplify the following expression.

निम्नलिखित व्यंजक को हल कीजिए।

$$\frac{(62\times 62\times 62)-3(62\times 62\times 22)+3(62\times 22\times 22)-(22\times 22\times 22)}{8\times 8\times 8}$$

HW

a) 225 b) 1250 c) 125 d) 25

a b

$$\frac{a^3-3a^2b+3ab^2-b^3}{8^3}=\frac{(a-b)^3}{8^3}$$

$62-22$

$$\frac{40^5}{8^3}=5^3=125$$

coaching center

(11) $\dfrac{5.169 \times 5.169 \times 5.169 - 64 \times (0.888)^3}{(5.169)^2 + (3.552)^2 + 4\,(0.888)(5.169)}$ is $(1+k)$, where $k =$ _____.

a 4^3 $4 \times .888$

$\dfrac{a^3 - b^3}{a^2+b^2+ab} = \dfrac{(a-b)(a^2+b^2+ab)}{a^2+b^2+ab}$

$1.617 = 1 + k$

0.323

b) 0.617

0.623

0.317

coaching center

(18) If $a^2 + b^2 + ab = 0$, then $(a^3 - b^3)$ is equal to

अगर $a^2 + b^2 + ab = 0$ है, तो $(a^3 - b^3)$:

a) 0 b) 1

c) $(a+b)^3$ d) $a^2b^4 + a^4b^2$

$a + \frac{1}{a} = \pm 1$

$a^3 = \pm 1$

$0 = (a-b)(a^2+ab+b^2)$ → 0

$\frac{a}{b} + \frac{b}{a} = -1$

$\Rightarrow \frac{a^3}{b^3} = 1$ $\quad a^3 = b^3$

e1 coaching center

(19) If $a^4 + b^4 = a^2b^2$, then $(a^6 + b^6)$ equals

अगर $a^4 + b^4 = a^2b^2$ है, तो $(a^6 + b^6)$:

a) 0 b) 1

c) $a^2 + b^2$ d) $a^2b^4 + a^4b^2$

$= a^2b^2$

$\frac{a^2}{b^2} + \frac{b^2}{a^2} = 1$

$\Rightarrow \frac{a^6}{b^6} = -1 \Rightarrow a^6 = -b^6$

$(a^2)^3 + (b^2)^3$

$0 = (a^2+b^2)(a^4+b^4-a^2b^2)$

0

coaching center

(20) If $\frac{a}{b}+\frac{b}{a}=1, a\neq 0, b\neq 0$ the value of a^3+b^3 is

HW अगर $\frac{a}{b}+\frac{b}{a}=1, a\neq 0, b\neq 0$ तो a^3+b^3 का मान:

a) 0 b) 1 c) -1 d) 2

(1) $a^2+b^2=ab$

$\Rightarrow a^2+b^2-ab=0$

$(a+b)(a^2+b^2-ab)=0$

$\Rightarrow 0$

---

(2) If $\frac{a}{b}+\frac{b}{a}=1$ then $\frac{a^3}{b^3}=-1$

$\Rightarrow a^3=-b^3 \Rightarrow a^3+b^3=0$

coaching center

(21) If $\frac{1}{x+y} = \frac{1}{x} + \frac{1}{y}$ then the value of $x^3 - y^3$ is

अगर $\frac{1}{x+y} = \frac{1}{x} + \frac{1}{y}$ तो $x^3 - y^3$ का मान:

a) 0 b) 1 c) -1 d) 2

$$\frac{1}{x+y} = \frac{x+y}{xy}$$

$$\Rightarrow xy = x^2 + y^2 + 2xy$$

$$\Rightarrow x^2 + y^2 + xy = 0$$

$$(x-y)(x^2+y^2+xy)$$

$$(x-y)(0)$$

coaching center

(22) If $a^3 + 3a^2 + 9a = 1$, then what is the value of $a^3 + \left(\frac{3}{a}\right)$?

यदि $a^3 + 3a^2 + 9a = 1$ हो, तो $a^3 + \left(\frac{3}{a}\right)$ का मान क्या है?

a) 31 b) 26 c) 28 d) 24

$3 \times (a^2 + 3a + 9) = \frac{1 \times 3}{a}$

$= 1 - 3a^2 - 9a + 3a^2 + 9a + 27$

$= 28$

$(a-3) \times (a^2 + 3a + 9) = \frac{1 \times (a-3)}{a}$

$\Rightarrow a^3 - 27 = 1 - \frac{3}{a}$

$\Rightarrow a^3 + \frac{3}{a} = 1 + 27 = 28$

(23) If $a + a^2 + a^3 - 1 = 0$, then what is the value of $a^3 + \frac{1}{a}$?

यदि $a + a^2 + a^3 - 1 = 0$ हो, तो $a^3 + \frac{1}{a}$ का मान क्या है?

a) 1 b) 4 c) 2 d) 3

$a + a^2 + a^3 = 1$

$\Rightarrow 1 + a + a^2 = \frac{1}{a}$

$= 1 - a - a^2 + 1 + a + a^2$

$= 2$

coaching center

(24) If $x^3 + \frac{3}{x} = 4(a^3 + b^3)$ and $3x + \frac{1}{x^3} = 4(a^3 - b^3)$, then $4(a^2 - b^2)$ is equal to:

यदि $x^3 + \frac{3}{x} = 4(a^3 + b^3)$ और $3x + \frac{1}{x^3} = 4(a^3 - b^3)$ है, तो $4(a^2 - b^2)$ बराबर है :

a) 0 b) 1 c) 2 d) 4

① ②

$4a^2 - 4b^2$

① + ② $\left(x + \frac{1}{x}\right)^3 = 8a^3 \Rightarrow x + \frac{1}{x} = 2a$

① − ② $\left(x - \frac{1}{x}\right)^3 = 8b^3 \Rightarrow x - \frac{1}{x} = 2b$

30 → ①

$4 \times x \times \frac{1}{x} = 4$

coaching center